



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
243 G.T. Road (N), Liluah, Howrah- 711204, West Bengal, India
 Ph: +91 33 26549315/17 Fax +91 33 26549318 Web: www.mckvie.edu.in/

Course Name:	Automotive Engines		
Course Code:	PC – AUE 501	Category:	Professional Core Courses
Semester:	Fifth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Engineering Thermodynamics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To familiarize with the terminology associated with IC engines		
2	To understand the basics of IC engines		
3	To understand combustion, and various parameters and variables affecting it in various types of IC engines		
4	To learn about various systems used in IC engines and the type of IC engine required for various applications		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: Basic air standard cycle Otto, Diesel & dual fuel cycle, comparison between Otto, Diesel and Dual fuel cycles. Basic Concepts- Air standard cycles and fuel-air cycles Assumptions Valve Timing diagram, Actual engine cycle.	3
2	Engine Construction: Construction and working of 4 stroke SI and CI Engine, Comparison between SI and CI engine, SI and CI engine fuel rating, octane number and cetane number, SI and CI Engine fuel properties, Alternative fuels (Alcohol, Biogas, Hydrogen, CNG, LPG).	3
3	SI Engine: Theory of Carburetion, Types of carburetors, Electronic fuel injection system, GDI. Combustion in spark Ignition engines, stages of combustion, flame propagation, rate of pressure rise, abnormal combustion. Phenomenon of Detonation in SI engines, effect of engine variables on Detonation. Combustion Chambers. Rating of fuels in SI engines and additives.	5
4	CI Engine: Fuel supply system, types of fuel pump, injector and distribution system, Combustion in compression ignition engines, stages of combustion, factors affecting combustion, Phenomenon of knocking in CI engine. Effect of knocking, Types of combustion chambers rating of fuels in CI engines. Additives Comparison of knocking in SI & CI engines, Concepts of Supercharging and Turbo charging.	5
5	Engine systems and components: Ignition system.(battery, magneto & electronic); Lubrication system; Engine starting system; Engine cooling system; Governing system (quality and quantity hit & miss governing); Intake and exhaust systems (two valves & four valves); Drive train (cam shaft, valves etc.). Variable valve timing, HCCI, PCCI engine	5

6	Fuels and Emissions: Chemical structure of the Petroleum, Refining process for petroleum, important qualities of the Engine fuels- (SI & CI engines), Diesel, and Gasoline fuels, Indian specifications. Alternate fuels (SI & CI engines)- Liquid fuels, gaseous fuels (LPG, NG, CNG), hydrogen and emulsified fuel. Air pollution due to IC engine, Engine emissions, Hydrocarbon emissions, (HC) & PM & Carbon monoxide emissions (CO), oxides of Nitrogen (NO _x) Euro norms , Bharat stage norms, Introduction to EDC and IDC, Introduction to carbon credit, Emission control methods for SI and CI engines, Electronic control module, Catalytic converters, EGR Concept of hybrid vehicles.	7
7	Cooling and Lubrication system: Need for cooling system. Types of cooling system, Liquid cooled system, Thermosyphon system, Pressure cooling system. Lubrication system, Mist lubrication system, Wet sump and dry sump lubrication. Properties of lubricants. Properties of coolants.	4
8	Performance characteristics & Testing of I.C. Engines: Introduction to Indian. Standards for testing of I.C. Engine, Mean effective pressure, indicated power, brake power, friction power, Methods to determine power and efficiencies Variables affecting performance of engine, characteristic curves, heat balance sheet, Methods of improving engine performance; super & turbocharged engines.	4
Total		36

Course Outcomes:

After completion of the course, students will be able to:

1	Know the basics of IC engines and the influence the different parameters the operational characteristics of IC Engines.
2	Understand the fundamentals of various automotive Engine and their construction details.
3	Understand the operation of automotive engine and importance of vehicle.

Learning Resources:

1	Ganesan V., Internal Combustion Engines, Tata McGraw Hill Co., Third Edition, 2007.
2	Obert E.F., Internal Combustion Engines and Air Pollution, Harper and Row Publication Inc. New York, 1973.
3	Heisler H., Advanced Engine Technology, Edward Arnold, 1995.
4	Heywood J.B., Internal Combustion Engine Fundamentals, McGraw Hill Book Co., New York, 1989.
5	Heldt P.M., High Speed Combustion Engines, Oxford & IBH publishing Co., India, 1985.
6	Stockel M.W., Stockel T.S. and Johanson C., Auto Fundamentals, The Goodheart, Wilcox Co. Inc., Illinois, 1996.



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Course Name:	Automotive Body and Chassis Engineering		
Course Code:	PC – AUE 502	Category:	Professional Core Courses
Semester:	Fifth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Engineering Mechanics, Theory of Machine
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To impart knowledge of different body construction and their sub assembly		
2	To identify automotive suspension components, their functions, and maintenance requirements		
3	To understand the differences between the linkage and rack and pinion type steering		
4	To understand the function, construction, and types of frames used on wheeled vehicles		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: Types of chassis layout with reference to power plant locations and drive, Vehicle frames. Various types of frames. Constructional details, Materials. Testing of vehicle frames. Unitized frame body construction: Loads acting on vehicle frame, chassis lubrication, and calculation of stresses on sections.	3
2	Front Axle and Steering System: Types of front axles. Construction details. Materials. Front wheel geometry viz. Castor, Camber, King pin inclination, Toe-in Conditions for true rolling motion of wheels during steering. Steering geometry. Ackerman and Davis steering system. Constructional details of steering linkages. Different types of steering gear boxes. Steering linkages and layouts. Power and power assisted steering.	4
3	Drive Line, Final Drive and Differential: Effect of driving thrust and torque reactions. Hotch kiss drive, torque tube drive and radius rods. Propeller shaft. Universal joints. Constant velocity universal joints. Front wheel drive. Different types of final drive. Differential principles. Construction and working of differential Non-slip differential. Differential locks.	3
4	Rear Axles: Construction of rear axles. Types of loads acting on rear axles. Full floating. Three quarter floating and semi floating rear axles. Rear axle housing. Construction of different types of axle housings. Multi-axled vehicles. Construction details of multi drive axle vehicles.	3
5	Suspension System: Need of suspension system- Types of front and rear suspension system- Suspension springs- Constructional details characteristics of leaf, coil and torsion bar springs- Independent suspension– Pneumatic suspension- constructional details of telescopic shock absorbers. Types, vibrations and riding comfort, role axis of spring suspension.).	4

6	Wheel & tyres: Types of wheels, construction, wired wheels, tyres, construction, types, radial, bias & belted bias, comparison, slip angle, under and over steering, tread patterns, tyre re-treading cold and hot, tyre specification tubeless tyre.	4
7	Braking System: Necessity of brake, stopping distance and time. Brake efficiency, weight transfer, brake shoe theory, determination of braking torque, braking systems- mechanical, hydraulic, disc, drum, parking and emergency brakes, power, servo and electrical brakes, details of hydraulic system, mechanical system and components. Types of master cylinders, Anti-lock braking systems.	4
8	Safety Aspect: Different Safety aspect, Driver's safety, passive restraint systems, Use of air bags, side impact analysis. Bumper system. Energy absorbent foams. Mechanisms in vehicle applied to safety.	3
9	Interior Ergonomics: Driver and passenger ergonomics with seating space arrangements for cars. Different types of car seats. Seat comfort, split frame seating, seat adjustment mechanisms. Visibility, methods of improving visibility.	4
10	Body Materials: Different types of ferrous and non-ferrous materials used in vehicle such as cast iron. Steel, Alloy steel, plastic, G.R.P, semi-rigid polyurethane.	2
11	Painting: Corrosion of vehicle body. Anticorrosion method. Paint and painting process.	2
Total		36

Course Outcomes:

After completion of the course, students will be able to:

1	Understand the operation of automotive systems and importance of vehicle frame.
2	Apply the knowledge of design consideration for braking system, suspension system and chassis.
3	Know different safety and ergonomic aspect to reduce human injury and increase comfort
4	Understand the fundamentals of various automotive body and their construction details.

Learning Resources:

1	Rjavee J.E., Automotive Technology- A System Approach, 3rd Edition, Thomson Asia Pte Ltd., Singapore, 2004.
2	De A., Automobile Engineering, Galgotia Publishers Pvt. Ltd., 2004.
3	Automobile Engineering, Ramalingam K.K., Scitech Publication (India) Pvt. Ltd., 2nd Edition, 2004.
4	Heitner J., Automotive Mechanics Principle and Practice, 2nd Edition, East West Press, 1999.
5	Powloski J., Vehicle Body Engineering, Business Books Ltd., London, 1989.
6	Fundamentals of Vehicle Dynamics, Thomas D. Gillespie Society of Automotive Engineers, 1992 ISBN 1560911999, 9781560911999



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Course Name:	Heat Transfer		
Course Code:	PC-AUE503	Category:	Professional Core Courses
Semester:	Fifth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Engineering Thermodynamics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To develop an understanding of Heat Transfer in Automobile Engineering		
2	To learn to use modes of heat transfer and to solve problems of engineering application		
3	To practice in the analytical formulation of heat transfer problems		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Introduction: Modes of heat transfer.	1
2	Conduction: Fourier law of heat conduction for isotropic material. Thermal conductivity. Derivation of the energy equation in three dimensions including transient effect. Non-dimensional - thermal diffusivity and Fourier number. One dimensional solution with and without heat generation in slab, cylinder and sphere. Analogy with electrical circuits. Critical thickness of insulation.	9
3	Fins: Heat flow through rectangular fin, Heat dissipation from an infinitely long fin, fin insulated at tip and fin losing heat at the tip, efficiency and effectiveness of fin, Weighted fin efficiency.	6
4	Conduction-Unsteady state: Lumped parameter approach, Biot number, Fourier number, Validity of lumped parameter approach. Introduction to Heissler Chart, time constant and response of temperature measuring instruments.	3
5	Radiation: Concept of radiation, absorptivity, reflectivity & transmissivity, black, white and grey surfaces, emissive power & emissivity. Intensity of radiation & solid angle, Laws of radiation – Planck's Law, Stefan-Boltzman Law, Wein's displacement Law, Kirchhoff's Law, Lambert's cosine law. Radiation exchange between black bodies and concept of shape factor. Radiation exchange between non-black Bodies, heat exchange between two grey surfaces, electrical analogy, radiation shield.	7
6	Convection: Introduction, Newton's law of cooling and significance of the heat transfer coefficient. Momentum and energy equations in two dimensions, importance of non-dimensional quantities and their physical significance. Order of magnitude analysis for flow over a flat plate. Velocity and thermal boundary layer thickness by integral method. Analogies between momentum, heat and mass transfer, Convective heat transfer for internal flow-concept.	5

7	Heat exchangers: Types, Heat exchanger analysis, overall heat transfer coefficient, fouling factor, LMTD for parallel & counter flow heat exchanger, effectiveness and number of transfer unit for parallel and counter flow heat exchanger, Radiator- correction factor for multi-pass arrangement.	5
Total		36

Course Outcomes:

After completion of the course, students will be able to:

1	Identify and formulate the heat conduction problems in rectangular, cylindrical & spherical co-ordinate system, using a mathematical model representing the physical system.
2	Understand and analyze the time dependent (transient) heat conduction
3	Apply the principles of convective heat transfer to compute heat transfer co-efficient in forced convection, natural convection for internal flows & external flows.
4	Understand the constructional features & working principles of heat exchangers which include the LMTD & NTU approach
5	Apply the radiation heat transfer problems between black & non-black bodies

Learning Resources:

1	R. K. Rajput., Heat and Mass Transfer, S. Chand ,2010
2	R.C. Sachdeba, Fundamental of Engineering Heat and Mass Transfer, New Age Science, 2010.
3	M. N. Ozisik, Heat Transfer - A Basic Approach, McGrawhill
4	J. P. Holman, Heat Transfer, 8th ed., McGrawhill
5	P.K. Nag, Heat & Mass Transfer, McGrawhill
6	Yunus Cengel, Heat and Mass Transfer: Fundamentals and Application, McGrawhill
7	Incropera and Dewitt, Fundamental of Heat and Mass Transfer, Wiley Publication



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Course Name:	Design of Machine Element		
Course Code:	PC-AUE504	Category:	Professional Core Courses
Semester:	Fifth	Credit:	4
L-T-P:	3-1-0	Pre-Requisites:	Engineering Mechanics and Strength of Materials
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			
1	To know mechanics of materials-based failure criteria underpinning the safety-critical design of machine components		
2	To learn overview of codes, standards and design guidelines for different elements		
3	To understand parameter optimization and design iteration		
4	To understand the relationships between component level design and overall machine system design and performance		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Design considerations- limits, fits and standardization, Review of failure theories for static and dynamic loading, Review and selection of materials and manufacturing processes.	6
2	Design of: Solid and hollow shafts, strength design of shafts, design based on torsional rigidity; Shaft coupling-rigid, pin-bush and geared flexible type, alignment of coupling.	9
3	Design of joints: Riveted Joints: Introduction, methods of riveting, Types of rivets, rivet materials, types of riveted joints, failures of riveted joints, joint efficiency. Welded Joints: Introduction, types of welded joints, design of welded joints (butt joints, fillet welds, axially loaded unsymmetrical welded joints, eccentrically loaded welded joints), threaded fasteners, pre-loaded bolts.	6
4	Introduction, types of springs, terminology, Helical compression spring - stress and deflection equations, stiffness, curvature effect: Wahl's factor, springs in parallel and series; Leaf springs, stresses in leaf springs.	3
5	Design of transmission elements: Gear materials, Beam Strength of Gear Teeth Lewis Equation, spur, helical, bevel and worm gears; belt and chain drives	8
6	Introduction, principle of hydro dynamic lubrication, bearing characteristic number and modulus, Sommerfeld number, coefficient of friction, design of journal bearings. Rolling contact bearings: types of bearings, static equivalent load, dynamic load rating, bearing life, selection of ball bearings.	8
7	Analysis of Clutches: Disc or Plate Clutch. multiple Disc, Cone Clutch. Brakes: Types, Energy Absorbed, Materials for Brake Lining, Internal Expanding Brake.	8
Total		48

Course Outcomes:	
After completion of the course, students will be able to:	
1	Design methodologies employed for the design of various machine components.
2	Formulate and analyze a design problem of machine parts.
3	Use of codes, standards and design guidelines.

Learning Resources:	
1	Shigley J.E. and Mischke C.R., Mechanical Engineering Design, 5th Edition, McGraw-Hill International, 1989.
2	Deutschman D., Michels W.J. and Wilson C.E., Machine Design Theory and Practice, Macmillan, 1992.
3	Juvinal R.C., Fundamentals of Machine Component Design, John Wiley, 1994.
4	Spottes M.F., Design of Machine Elements, Prentice-Hall India, 1994.
5	Norton R.L., Mechanical Design – An Integrated Approach, Prentice Hall, 1998
6	V. B. Bhandari, Design of Machine Elements by, McGraw Hill Publishing Co. Ltd., 2007.



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Course Name:	Economics for Engineers		
Course Code:	HM-HU 501	Category:	Management Science and Humanities
Semester:	Fifth	Credit:	3
L-T-P:	3-0-0	Pre-Requisites:	Mathematics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 5

Course Objectives:	
1	To understand the role and scope of Engineering Economics and the process of economic decision making along with the different concepts of cost and cost estimation techniques
2	To familiarize with the concepts of cash flow, time value of money and different interest formulas
3	To understand the role of uncertainty in future events and using different concepts from probability to deal with uncertainty
4	To understand the concepts of Depreciation and Replacement analysis along with their methods of calculation and familiarization with the phenomenon of inflation and the use of price indices in engineering Economics
5	To know the basic concepts of Accounting and Financial Management

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Economic Decisions Making: Overview, Problems, Role, Decision making process. Engineering Costs & Estimation – Fixed, Variable, Marginal & Average Costs, Sunk Costs, Opportunity Costs, Recurring And Nonrecurring Costs, Incremental Costs, Cash Costs vs Book Costs, Life-Cycle Costs; Types Of Estimate, Estimating Models - PerUnit Model, Segmenting Model, Cost Indexes, Power-Sizing Model, Improvement & Learning Curve, Benefits.	9
2	Cash Flow, Interest and Equivalence: Cash Flow – Diagrams, Categories & Computation, Time Value of Money, Debt repayment, Nominal& Effective Interest. Cash Flow & Rate of Return Analysis – Calculations, Treatment of Salvage Value, Annual Cash Flow Analysis, Analysis Periods; Internal Rate of Return, Calculating Rate of Return, Incremental Analysis; Best Alternative Choosing an Analysis Method, Future Worth Analysis, Benefit-Cost Ratio Analysis, Sensitivity and Breakeven Analysis. Economic Analysis in The Public Sector -Quantifying And Valuing Benefits & drawbacks.	9
3	Inflation and Price Change – Definition, Effects, Causes, Price Change with Indexes, Types of Index, Composite vs Commodity Indexes, Use of Price Indexes In Engineering Economic Analysis, Cash Flows that inflate at different Rates.	9

	Present Worth Analysis: End-Of-Year Convention, Viewpoint Of Economic Analysis Studies, Borrowed Money Viewpoint, Effect Of Inflation & Deflation, Taxes, Economic Criteria, Applying Present Worth Techniques, Multiple Alternatives. Uncertainty In Future Events - Estimates and Their Use in Economic Analysis, Range Of Estimates, Probability, Joint Probability Distributions, Expected Value, Economic Decision Trees, Risk, Riskvs Return, Simulation, Real Options.	
4	Depreciation: Basic Aspects, Deterioration & Obsolescence, Depreciation and Expenses, Types Of Property, Depreciation Calculation Fundamentals, Depreciation And Capital Allowance Methods, Straight-Line Depreciation Declining Balance Depreciation, Common Elements Of Tax Regulations For Depreciation And Capital Allowances. Replacement Analysis - Replacement Analysis Decision Map, Minimum Cost Life of a New Asset, Marginal Cost, Minimum Cost Life Problems. Accounting – Function, Balance Sheet, Income Statement, Financial Ratios Capital Transactions, Cost Accounting, Direct and Indirect Costs, Indirect Cost Allocation.	9
Total		36

Course Outcomes:

After completion of the course, students will be able to:

1	Discuss fundamentals of economic analysis.
2	Describe rate of return and profitability analysis, Present, Future, Annuity, Risk and return, BEP and Sensitivity Analysis, Bayesian joint probability and quantitative decision making, basic accounting system and balance sheet and P & L accounts etc.
3	Apply decision making skills in terms of Economic, financial considerations in practice.
4	Apply knowledge to take right financial decision at the right point in time in real world situation.

Learning Resources:

1	James L.Riggs, David D. Bedworth, Sabah U. Randhawa : Economics for Engineers 4e, Tata McGraw-Hill
2	Donald Newnan, Ted Eschembach, Jerome Lavelle: Engineering Economics Analysis, OUP
3	R.PaneerSeelvan: Engineering Economics, PHI
4	Sullivan and Wicks: Engineering Economy, Pearson
5	John A. White, Kenneth E. Case, David B. Pratt : Principle of Engineering Economic Analysis, John Wiley



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Course Name:	Fluid Mechanics and Heat Transfer Lab		
Course Code:	PC-AUE591	Category:	Professional Core Courses
Semester:	Fifth	Credit:	1.5
L-T-P:	0-0-3	Pre-Requisites:	Engineering Thermodynamics, Fluid Mechanics and Hydraulic Machines, Heat Transfer
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05
Course Objectives:			
1	To understand the principles and performance characteristics of flow and thermal devices		
2	To know about the measurement of the fluid and thermal properties		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Measurement of co-efficient of discharge of given orifice and venturimeters.	3
2	Determination of the co-efficient of friction factor for flow through pipes.	3
3	Determination of the performance characteristics of a centrifugal pump.	3
4	Determination of the performance characteristics of Pelton Wheel	3
5	Determine the flow rate and velocity profile in a duct using pitot tube.	3
6	Determination of thermal conductivity of a metal rod and/or insulating powder materials	3
7	Heat transfer through forced convection.	3
8	Heat transfer through natural convection from a vertical surface.	3
9	Determination of the convective heat transfer coefficient for flow over a heated plate	3
10	Measurement of emissivity in a test surface.	3
11	Experiment with a parallel flow and a counter flow heat exchanger.	3
12	Heat transfer through a pin fin.	3
13	Study of Basic hydraulic circuit for the working of double acting cylinder and a hydraulic motor.	3
Total (About 12 experiments will be carried out as listed above)		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Measure various properties of thermal and fluids.
2	Characterize the performance of fluid/ thermal machinery

Learning Resources:	
1	A Textbook of Fluid Mechanics and Hydraulic Machines in SI Unit, R.K.Bansal, Laxmi Publication (P) Ltd., 10 th Edition, 2018
2	Heat and Mass Transfer by R. K. Rajput., S. Chand ,2010



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Course Name:	Automobile Engineering Lab I (Engine and Chassis Component Lab)		
Course Code:	PC-AUE592	Category:	Professional Core courses
Semester:	Fifth	Credit:	1.5
L-T-P:	0-0-3	Pre-Requisites:	Engineering Thermodynamics and Engineering Mechanics
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05
Course Objectives:			
1	To train the Students to know the details of different automobile components.		
2	To dismantling, assembling, inspect, measures and testing of components.		

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	Dismantling, measurement, inspection and assembling of different modern engine [like Multipoint fuel injection (MPFI) and Common rail injection (CRI) engines and Digital twin spark ignition (DTSI) etc.] engine for passenger car, commercial vehicle and two wheeler engines.	3
2	Study of fuel supply system (SI and CI) and structure and testing of common rail high pressure injectors.	3
3	Dismantling, assembling and testing of different types of Fuel injection Pumps such as distributor type, high pressure pump.	3
4	Electronic ignition and battery ignition system with accessories.	3
5	Study of cooling, lubrication.	3
6	Study and testing of automotive air conditioning system.	3
7	Dismantling and assembling of different types of clutch.	3
8	Dismantling and assembling of different types of Gear.	3
9	Dismantling and assembling of different Steering system and study of driver seat.	3
10	Study of Frames used for Heavy commercial vehicle (HCV), Car, Two & Three Wheelers and Dismantling and assembling of Suspension system.	3
11	Dismantling and assembling of Braking system, Brake adjustment and brake bleeding.	3
12	Dismantling and assembling of Wheels and Tyres. Dismantling and assembling of Propeller Shaft, Universal Joints and Differential.	3
Total		36

Course Outcomes:

After completion of the course, students will be able to:

1	Ability to dismantle, assemble, inspect, measure and testing the automobile engine components.
2	This Laboratory course is intended to give the students, experimental knowledge about various automotive chassis components.

Learning Resources:

1	Automobile Engineering, Ramalingam K.K., Scitech Publication (India) Pvt. Ltd., 2nd Edition, 2004.
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Course Name:	Soft Skill Development Lab		
Course Code:	HM-HU591	Category:	Humanities and Social Sciences including Management Courses
Semester:	Fifth	Credit:	1
L-T-P:	0-0-2	Pre-Requisites:	Students must have basic knowledge of English Language
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05
Course Objectives:			
1	To equip the students with good communication skills		
2	To enable the students to think and speak effectively on everyday topics, including topics related to technical concepts		
3	To prepare them for interviews and future job environments		
4	To develop industry-ready attitude towards professional communication		

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Conversation Practice Sessions - General Conversation - Warm-up sessions - Basics of Communication, verbal and non-verbal communication.	6
2	Group Discussion - Group Discussion & Debates, Do's & Don'ts, etc., Intensive Practice Sessions.	12
3	Interview sessions: Principles and practices of Personal Interview • Do's and Don'ts of facing an interview. • SWOC Analysis • Rigorous practices of mock-interviews.	9
4	Presentations: Fundamentals of presentation skills, Secrets of an effective presentation, Presentation Practice Sessions with the help of power point presentation and other audio-visual aids, Face question answer sessions at the end of their presentation.	9
Total		36

Course Outcomes:	
After completion of the course, students will be able to:	
1	Acquire over all Communicative Competence.
2	Develop Team Building and Leadership Quality.
3	Deliver an enthusiastic and well-practiced presentation
4	Communicate with clarity and confidence thereby enhancing employability skills of the students.

Learning Resources:	
1	Meenakshi Raman and Shalini Upadhyay, Soft Skills: Key to success in Workplace and Life,
2	Sanjay Kumar and Pushp Lata, Communication Skills., Oxford University Press, 2011.
3	Monipally: Business Communication, Tata McGraw Hill
4	Madhukar: Business Communications; Vikas Publishing House
5	Senguin J: Business Communication; Allied Publishers
6	Business Communication: Rajendrapal & Korlahalli



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Course Name:	Project-I (Mini Project)		
Course Code:	PW-AUE581	Category:	Project/ Internships/ Sessional
Semester:	Fifth	Credit:	1
L-T-P:	0-0-2	Pre-Requisites:	Nil
Full Marks:	100		
Examination Scheme:	Semester Examination: 100		
Course Objectives:			
1	To acquaint with the process of undertaking literature survey/industrial visit and identifying the problem		
2	To familiarize the process of solving the problem in a group		
3	To acquaint with the process of applying basic engineering fundamental in the domain of practical application		
4	To inculcate the process of research		

Course Contents:	
Description of Topic	Contact Hrs.
The project work could be done in the form of a summer project or internship in the industry or even a minor practical project in the college. Participation in any technical event/ competition to fabricate and demonstrate an innovative machine or product could be encouraged under this course.	30
Total	30
Course Outcomes:	
After completion of the course, students will be able to:	
1	Fulfill team roles assigned & communicate effectively.
2	Solve engineering problems involving current issues using modern tools.
3	Demonstrate the ability to apply the knowledge gained in the programme.



MCKV INSTITUTE OF ENGINEERING

NAAC Accredited "A" Grade Autonomous Institute under UGC Act 1956
 Approved by AICTE & affiliated to Maulana Abul Kalam Azad University of Technology, West Bengal
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Course Name:	Aptitude Skill Development-I		
Course Code:	MC571	Category:	Mandatory Courses
Semester:	Fifth	Credit:	0
L-T-P:	2-0-0	Pre-Requisites:	Basic knowledge of Mathematics and English Language
Full Marks:	100		
Examination Scheme:	Semester Examination: 100		

Course Objectives:	
1	To be familiar with the basic concepts of QUANTITATIVE ABILITY.
2	To be familiar with the basic concepts of LOGICAL REASONING Skills .
3	To be familiar with the basic concepts of PROBABILITY.
4	Acquire knowledge in VERBAL REASONING and VOCABULARY

Course Contents:		
Module No.	Description of Topic	Contact Hrs.
1	Basics of Quantitative Abilities: Number System, HCF and LCM, Average, Ratio, Proportion and Variations, Problems on Percentage.	4
2	Arithmetic Quantitative Abilities: Problems on Ages, Profit and Loss, Time and Work, Problems on Simple and Compound Interest, Problems on Time, Speed and Distance.	6
3	Permutation and Combination, Set theory, Venn Diagram, Probability	5
4	Logical Reasoning: Number Series, Alpha Numerical, Letter & Symbol Series, Syllogisms Numerical and Alphabet Puzzles, Seating Arrangement, Blood Relation and Calendars.	7
5	Data Interpretation	2
6	Verbal: Analogies, Antonym, Synonym, Sentence Correction, Fill in the Blanks	3
Total		27

Course Outcomes:

After completion of the course, students will be able to:

1	Understand the basic concepts of QUANTITATIVE ABILITY.
2	Understand the basic concepts of LOGICAL REASONING Skills .
3	Understand the basic concepts of PROBABILITY.
4	Acquire satisfactory competency in use of VERBAL REASONING

Learning Resources:

1	Arun Sharma, “Quantitative abilities”, McGraw-Hill
2	R.S.Agrawal, “Quantitative Aptitude for Competitive Examinations”, S. Chand
3	R.S.Agarwal,”A Modern Approach to Verbal & Non-Verbal Reasoning ”,S.Chand